

This hypothetical scenario illustrates the need for a minimum pricing increment that is not too small.

Too big of a minimum pricing increment is also problematic since it would reduce the quality of price discovery by precluding price competition for providing liquidity.²⁰ More specifically, too large a tick size can increase transaction costs for investors by artificially widening the “bid-ask spread” – the difference between the bid (highest price a buyer is willing to pay) and the ask (the lowest price a seller is willing to accept) prices.²¹ For example, consider a hypothetical scenario where a liquidity provider is willing to bid \$10.121 to buy a stock and offer \$10.124 to sell the stock. If the tick size were \$0.005, the resulting bid and offer from this liquidity provider would be \$10.120 and \$10.125, respectively, with a spread of \$0.005. If the tick size were \$0.01, the corresponding bid and offer would be \$10.120 and \$10.130, with a spread of \$0.01.²² In other words, but for the requirement under Rule 612 that sets the tick size to be \$0.01 for quotes and orders in NMS stocks priced at or above \$1.00, a smaller tick size would have narrowed spreads in some instances and allowed prices to better reflect the underlying economics for certain NMS stocks. As explained below, up to 74.3% of the share volume transacted in NMS stocks in 2023 may have bid-ask spreads that are constrained by the current minimum pricing increments.²³ These widened bid-ask spreads increase transaction costs for

²⁰ See infra section VII.B.2.

²¹ See infra section VII.B.2; see also 17 CFR 242.600(b)(16).

²² See infra section VII.B.2 (providing a similar example showing how a minimum pricing increment could double the width of a bid-ask spread).

²³ See infra section VII.C.1.b (discussing percentage of share volume likely to be tick-constrained). See also infra section VII.B.2 (discussing the definition of “tick-constrained”).

investors.²⁴ Conversely, a smaller tick size that allows for narrower bid-ask spreads would benefit investors by reducing transaction costs.²⁵

The minimum pricing increments in Rule 612 were adopted in 2005, when the Commission adopted Regulation NMS, and it was an adjustment in a long series of adjustments to the minimum pricing increments over time. For many decades, the U.S. equity markets used fractions of a dollar as minimum pricing increments (e.g., 1/8, 1/16, and 1/32 of a dollar).²⁶ Prior to 1997, the minimum pricing increment on the New York Stock Exchange LLC (“NYSE”) for stocks above \$1.00 per share was 1/8 of a dollar (or 12.5 cents).²⁷ In 1997, NYSE and the Nasdaq Stock Market LLC (“Nasdaq”) revised their rules to use the minimum pricing increment of 1/16 of a dollar (6.25 cents).²⁸ In January 2000, the Commission mandated decimal pricing (i.e., moving from fractional increments to penny increments) in certain securities,²⁹ and by April 2001, the market had fully converted to decimal pricing.³⁰

Up to this point, minimum pricing increments for NMS stocks were set by the individual trading venues. But in 2004, as part of Regulation NMS and pursuant to the authority under the

²⁴ See infra section VII.B.2.

²⁵ See infra section VII.B.2.

²⁶ See Staff Report to Congress on Decimalization, Commission (July 2012) (“Staff Decimalization Report”), available at <https://www.sec.gov/files/decimalization-072012.pdf>, at 4. Staff reports, Investor Bulletins, and other staff documents (included those cited herein) represent the views of Commission staff and are not a rule, regulation, or statement of the Commission. The Commission has neither approved nor disapproved the content of these staff documents, and, like all staff documents, they have no legal force or effect, do not alter or amend the applicable law, and create no new or additional obligations for any person.

²⁷ See Self-Regulatory Organizations; New York Stock Exchange, Inc.; Order Granting Approval to Proposed Rule Change Relating to Trading Differentials for Equity Securities, 62 FR 42847, 42848 n.5 (Aug. 8, 1997). See also Division of Market Regulation, Market 2000: An Examination of Current Equity Market Developments (1994), available at <https://www.sec.gov/divisions/marketreg/market2000.pdf>, at 37-38, fn. 43 (describing NYSE’s tick size of 1/8 of a dollar in 1994).

²⁸ See Staff Decimalization Report, supra note 26, at 4-5.

²⁹ See Securities Exchange Act Release No. 42360 (Jan. 28, 2000), 65 FR 5003 (Feb. 2, 2000).

³⁰ See Staff Decimalization Report, supra note 26, at 5-6.

1975 Amendments, the Commission proposed Rule 612 to implement market-wide uniform minimum pricing increments for quoting in NMS stocks.³¹ The Commission stated that, while the benefits of decimal pricing had justified the costs, there was a potential for costs to investors and the markets to surpass the benefits if the minimum pricing increment decreased beyond a certain level, and the proposed rule was designed to address the scenario where market participants attempt to step ahead of competing limit orders at the smallest economic increment possible.³² Thus, the Commission adopted Rule 612 in 2005, which established the minimum pricing increments of \$0.01 for quotations and orders of NMS stocks priced at, or greater than, \$1.00 per share, and \$0.0001 for quotations and orders of NMS stocks priced under \$1.00 per share. The Commission stated that, at the time, it did not believe that the potential benefits of marginally better prices offered by allowing sub-penny quoting in securities were likely to justify the costs of permitting such quotes.³³

When the Commission adopted Rule 612 in 2005, it acknowledged that the markets could evolve over time and shift the balance of the costs and benefits of the adopted tick size.³⁴ Two decades later, the market has evolved considerably, and amendments to Rule 612 are necessary to continue to further the objectives of the Exchange Act. Data analysis shows that stocks with

³¹ See Securities Exchange Act Release No. 49325 (Feb. 26, 2004), 69 FR 11126, 11171 (Mar. 9, 2004) (“2004 Regulation NMS Proposing Release”) (“the Commission is proposing a rule that would prohibit every national securities exchange, national securities association, ATS (including ECNs), vendor, broker or dealer from ranking, displaying, or accepting from any person a bid or offer, an order, or an indication of interest in any NMS stock in an increment less than \$0.01.”).

³² See Regulation NMS Adopting Release, supra note 4, at 37551-52 (citing 2004 Regulation NMS Proposing Release at 11165).

³³ See Regulation NMS Adopting Release, supra note 4, at 37553 (“Even assuming that quoting in sub-penny increments would reduce spreads, the Commission continues to believe, on balance, that the costs of sub-penny quoting are not justified by the benefits.”).

³⁴ Id. (“Nevertheless, the Commission acknowledges the possibility that the balance of costs and benefits could shift in a limited number of cases or as the markets continue to evolve.”).

sufficiently narrow bid-ask spreads would trade better, namely it would be easier and less costly for investors to transact, if they were allowed to quote at increments smaller than one penny.³⁵ Indeed, for these stocks, the risks of “stepping ahead” are lowered while the benefits of greater price competition from relaxing the “tick constraint” are greater.³⁶

2. Proposed and Adopted Amendments

Accordingly, the Commission proposed amendments to Rule 612 to introduce three minimum pricing increments that were less than \$0.01 (i.e., \$0.005, \$0.002, \$0.001) for quotes and orders priced \$1.00 or more for certain NMS stocks based upon each stock’s time weighted average quoted spread (“TWAQS”).³⁷ The proposed amendments would have assigned sub-penny minimum pricing increments to any NMS stock that had a TWAQS of \$0.04 or less. This proposed amendment was designed to address the issues related to tick-constrained stocks described above that have arisen since 2005. The Commission also proposed to impose these minimum pricing increments for trades, subject to certain exceptions.

As explained below, in response to commenters, the Commission is adopting modified amendments to Rule 612 to introduce one minimum pricing increment that is less than \$0.01, i.e., \$0.005, for quotes and orders priced \$1.00 or more for NMS stocks that have a TWAQS of \$0.015 or less.³⁸ The Commission is not adopting a minimum pricing increment for trades.³⁹

³⁵ See infra section VII.D.1.

³⁶ See infra sections VII.B.2 and VII.D.1.b.

³⁷ See Proposing Release, supra note 11, at 80280.

³⁸ See infra section III.C.

³⁹ See infra section III.D.

B. Rule 610 Fees for Access to Quotations and Transparency of Fees

1. Background

Trading centers⁴⁰ can choose to charge an access fee, or pay a rebate, to the participants – liquidity providers (market participants with orders resting at the trading center) and liquidity takers (market participants who submit incoming orders to execute against orders resting at the trading center) – who trade at their venue. As discussed in section VII.C.2.b, the predominant exchange fee structure is maker-taker, in which an exchange charges a fee to liquidity takers and pays a rebate to liquidity providers, and the rebate is typically funded through the access fee.⁴¹

As adopted in 2005, Rule 610(c) set the access fee cap for protected quotations⁴² priced at \$1 or more at 30 cents per 100 shares (“30 mils” per share) for NMS stocks. Rule 610(c) also applies to any other quotation of a trading center that is the best bid or offer of an exchange or association.⁴³ The access fee cap was based, in part, upon the prevailing fees that were charged by certain trading centers at that time.⁴⁴ For NMS stocks priced below \$1, the fee cap was set at 0.3% of the quotation price.⁴⁵ Rule 610 was adopted at the same time as Rule 611, the Order

⁴⁰ 17 CFR 242.600(b)(106) (providing a definition of the term “trading center”). This discussion focuses on exchange fees because, currently, exchanges are the only trading centers that have quotations that are subject to the access fee caps under Rule 610(c). See infra note 367.

⁴¹ See also infra sections VII.B.3 and VII.C.2.c, Table 5 and Table 6 (showing the predominance of both dollar and share exchange trading volume occurs on maker-taker venues).

⁴² 17 CFR 242.610(c). A protected quotation is defined in Rule 600(b)(82) as “a protected bid or protected offer.” 17 CFR 242.600(b)(82). A protected bid or protected offer is defined as “a quotation in an NMS stock that: (i) Is displayed by an automated trading center; (ii) Is disseminated pursuant to an effective national market system plan; and (iii) Is an automated quotation that is the best bid or best offer of a national securities exchange, or the best bid or best offer of a national securities association.” 17 CFR 242.600(b)(81).

⁴³ For purposes of this discussion, references to protected quotations under Rule 610(c) also include manual quotations that are the best bid or best offer of an exchange or association.

⁴⁴ See Regulation NMS Adopting Release, supra note 4, at 37545.

⁴⁵ See Regulation NMS Adopting Release, supra note 4, at 37544 n.406.

Protection Rule, which established intermarket protection against trade-throughs⁴⁶ for all NMS stocks. Rule 610(c) was designed to preclude trading centers that posted protected quotations from raising their fees in an attempt to take improper advantage of the trade-through protections adopted under Rule 611.⁴⁷ The Commission designed the access fee caps to preserve the benefits of both the strengthened price protection under Rule 611 and the more efficient linkages among trading centers that were developed under Regulation NMS to access protected quotations because the benefits could be compromised if substantial fees were charged.⁴⁸

Since an access fee that is too high when compared to the tick size can create pricing distortions, the access fee caps need to be adjusted in conjunction with the reduction in tick size to prevent such distortions.⁴⁹ In addition, as discussed below, many exchanges charge the maximum fee allowed to access protected quotes, and primarily use those fees to pay rebates to market participants that provide liquidity.⁵⁰ This practice raises a number of concerns and may interfere with section 11A's objectives of ensuring the fairness and usefulness of quotation information.⁵¹

First, the actual prices, inclusive of fees and rebates, for investors and other market participants to trade a stock are not fully transparent. In general, the higher the permitted level of

⁴⁶ A trade-through occurs when a trading center executes an order at a price that is inferior to the price of a protected quotation that is displayed by another trading center. See 17 CFR 242.600(b)(105) for the definition of trade-through under Regulation NMS.

⁴⁷ See Regulation NMS Adopting Release, supra note 4, at 37544 and 37595.

⁴⁸ See Regulation NMS Adopting Release, supra note 4, at 37544.

⁴⁹ See infra sections IV.D.1 and VII.D.2.a. See also Proposing Release, supra note 11, at 80348 (stating "the access fee cap should not be greater than ½ of the tick size in order to preserve coherence between net and nominal price rankings of trading venues.").

⁵⁰ See infra sections VII.B.3 and VII.C.2.

⁵¹ See infra sections IV.D.1 and VII.B.3. See also Regulation NMS Adopting Release, supra note 4, at 37545 ("For quotations to be fair and useful, there must be some limit on the extent to which the true price for those who access quotations can vary from the displayed price.").

access fees, the higher the rebates, and the greater the potential discrepancy between displayed quoted prices on the one hand, and actual prices on the other.⁵²

Furthermore, exchanges' use of fees and rebates creates a potential conflict of interest between broker-dealers and their customers with respect to broker-dealer order routing, by providing incentives for a broker-dealer to route customer orders to certain exchanges to receive higher rebates or avoid higher fees based on their own economic interest.⁵³ This potential conflict of interest is exacerbated if broker-dealers do not fully pass on the fees and rebate to their customers, since rebate-seeking by broker-dealers may come at the cost of execution quality of customers.⁵⁴ In addition, exchanges use complex fee schedules. Generally, the higher the access fee cap, the wider the range of possible fees and rebates, which results in more complex pricing schedules. Such complexity makes it more costly for market participants to design and implement order execution strategies.

Finally, exchanges' fee and rebate schedules are typically calculated at month's end, which requires market participants to make trading decisions without the ability to determine their full trading costs at the time of execution.⁵⁵ In turn, this lack of transparency impedes a market participant's ability to evaluate fully where to send its orders because the market participant cannot calculate the fees and rebates that will apply to the order contemporaneous with execution.⁵⁶ Concerns with such lack of price transparency are exacerbated when various

⁵² See infra sections IV.B.2, VII.D.2, and VII.E.1. In certain cases, the disparity between market quotations and actual transaction costs may be substantial. See, e.g., Proposing Release, supra note 11, at 80328.

⁵³ See infra sections IV.B.2, IV.D and VII.D.3.

⁵⁴ See text accompanying infra note 1518.

⁵⁵ See infra sections IV.E, VII.C.2, and VII.D.3.

⁵⁶ See infra sections IV.E, VII.C.2, and VII.D.3.

exchanges have different fee schedules, as it is difficult for market participants to compare net prices across markets.

2. Proposed and Adopted Amendments

Accordingly, the Commission proposed to amend Rule 610 in two ways. First, to accommodate the proposed smaller minimum pricing increments under proposed Rule 612, as well as to address the distortions that have developed under the access fee caps, the Commission proposed to reduce Rule 610(c)'s 30 mil cap for executions against protected quotations priced \$1.00 or more as follows: a \$0.001 (or 10 mils) access fee cap for NMS stocks that would have been assigned a minimum pricing increment larger than \$0.001; and a \$0.0005 (or 5 mils) access fee cap for NMS stocks that would have been assigned a \$0.001 minimum pricing increment. For protected quotations in NMS stocks priced under \$1.00 per share, the Commission proposed to reduce the 0.3% fee cap to 0.05% of the quotation price.

As discussed in detail below, in response to comments, the Commission is adopting amendments to Rule 610(c) with modifications from the proposal. Specifically, in light of the amendments to Rule 612, the Commission is adopting only the proposed 10 mil per share access fee cap for all protected quotations priced \$1.00 or more.⁵⁷ For protected quotations priced less than \$1.00, the Commission is adopting an access fee cap of 0.1% of the quotation price per share.⁵⁸ As discussed in section VII.D.2.b, the adopted amendments to the access fee caps will not impede the ability of exchanges to fund their execution services.

Second, to facilitate the ability of market participants to understand and calculate the total price of transactions at the time of execution, the Commission proposed an amendment to Rule

⁵⁷ See infra section IV.D.

⁵⁸ See id.

610 to add subpart (d) to require that all exchange fees charged, and rebates paid, for the execution of an order in an NMS stock be determinable at the time of execution. As discussed in detail below, the Commission is adopting Rule 610(d) as proposed.

C. Transparency of Better Priced Orders

1. Background

The widespread availability of timely information with respect to quotations for and transactions in NMS stocks (“NMS information”) is critical to the ability of market participants to participate effectively in the U.S. securities markets.⁵⁹ NMS information is currently disseminated within the national market system by the exclusive plan processors (“exclusive securities information processors” or “SIPs”).⁶⁰

In 2020, the Commission adopted amendments to Regulation NMS to modernize the NMS information provided within the national market system for the benefit of market participants and to better achieve section 11A’s goals of assuring “the availability to brokers, dealers, and investors of information with respect to quotations for and transactions in securities that is prompt, accurate, reliable, and fair” (“MDI Rules”).⁶¹ In light of delays in the

⁵⁹ NMS information is made widely available to investors through the national market system and “serves an essential linkage function by helping to assure that the public is aware of the best displayed prices for a stock, no matter where they may arise in the national market system.” See Securities Exchange Act Release No. 61358 (Jan. 14, 2010), 75 FR 3594 (Jan. 21, 2010) (“Concept Release on Equity Market Structure”) at 3600. The availability of NMS information also “enables investors to monitor the prices at which their orders are executed and assess whether their orders received best execution.” Id.

⁶⁰ There are three effective national market system plans that govern the collection, consolidation, processing and dissemination of quotation and transaction information for NMS stocks: the Consolidated Tape Association Plan (“CTA Plan”); the Consolidated Quotation Plan (“CQ Plan”); and the Joint Self-Regulatory Organization Plan Governing the Collection, Consolidation, and Dissemination of Quotation and Transaction Information for Nasdaq-Listed Securities Traded on Exchanges on an Unlisted Trading Privileges Basis (“UTP Plan”) (together the “Equity Data Plans”). Currently, the Securities Industry Automation Corporation (“SIAC,” an affiliate of the NYSE) is the exclusive SIP for the CTA and CQ Plans, and Nasdaq is the exclusive SIP for the UTP Plan. See MDI Adopting Release, supra note 10, at 18728. Each exclusive SIP is the plan processor for one of the Equity Data Plans.

⁶¹ See MDI Adopting Release, supra note 10.

implementation of the MDI Rules, the Commission is accelerating the implementation of the round lot and odd-lot information definitions adopted as part of the MDI Rules so that investors will benefit sooner from greater transparency and accessibility of better priced orders⁶² and improved ability to assess the execution quality of their orders, as explained below.⁶³

Until the full implementation of the MDI Rules, NMS information disseminated within the national market system by the exclusive SIPs includes, for each NMS stock, the price, size, and exchange of each last sale, each exchange's current highest bid and lowest offer and the shares available at those prices (the best bid and best offer or "BBO"), the national best bid and national best offer ("NBBO"), odd-lot⁶⁴ transaction information, and certain regulatory and administrative data ("SIP data").⁶⁵ Information on NMS stock quotations is provided in round lots, and, until the round lot definition adopted in the MDI Rules is implemented, round lots are defined in rules of the exchanges.⁶⁶ For most NMS stocks, exchange rules define a round lot as

⁶² "Better priced orders" refers to orders that are priced superior to the national best bid and national best offer but are not included in NMS information because they consist of too few shares. See infra notes 66-68 and accompanying text. The MDI Rules' round lot and odd-lot information definitions will allow better priced orders to be included in NMS information so that market participants that subscribe to the exclusive SIP feeds (that otherwise would not be able to view these orders without purchasing exchange proprietary feeds) will be able to view and access these orders.

⁶³ See infra sections V.C.1.a and VII.D.4.

⁶⁴ Odd-lot is defined in Rule 600(b)(68) as an order for the purchase or sale of an NMS stock in an amount less than a round lot. 17 CFR 242.600(b)(68).

⁶⁵ See Proposing Release, supra note 11, at 80294. Under the decentralized consolidation model established by the MDI Rules, NMS information will consist of "consolidated market data," as defined in the MDI Rules. 17 CFR 242.600(b)(24).

⁶⁶ See Proposing Release, supra note 11, at 80294 n.328. A "round lot" is not defined in the Exchange Act and, prior to the MDI Rules, it was not defined in Regulation NMS. Exchange rules typically define a round lot as 100 shares, but they also allow the exchange, or the primary listing exchange for the stock, discretion to define it otherwise. See, e.g., NYSE Rule 7.5 ("A 'round lot' is 100 shares, unless specified by the primary listing market to be fewer than 100 shares.").